

be a more efficient and secure online transaction method that works within traditional financial systems at a lower transaction cost to users.

## **Litecoin**

Litecoin is the third largest open source digital peer-to-peer cryptocurrency in the world based on the market capitalisation of USD \$586,759,042. It is built on the Bitcoin protocol - designed to be decentralised and has no central operator. The goal of its design is to upgrade some features that are thought to be lacking in the Bitcoin system and presents itself as a viable alternative to Bitcoin. Just like Bitcoin it is divisible to 100,000,000 units with LTC as its signifier.

The main differences from Bitcoin include faster transaction confirmations. As we know the Bitcoin protocol takes 10 minutes to process a block which needs to be confirmed in the ledger. Litecoin processes blocks every 2.5 minutes allowing for faster confirmations. This also increases the network's security against attacks for double spending. Litecoin is also based on a different proof-of-work system using script. This key difference is a means to deny miners with GPU, FPGA and ASIC systems an advantage over regular CPU owners. Normally it is harder for ASIC and FPGA systems to work with script making the process of mining using these hardware is more expensive. However this claim is under debate with evidence indicating that the result is in fact the opposite due to its poor use of script.

The goal behind the intended difficulty is to ensure that a select group of individuals who can afford high end computing power do not unfairly gain most of the Litecoins that are mined. This system is intended to make small scale miners achieve a level playing field.

The Litecoin system is designed to produce just under 84 million litecoins and it was valued at USD \$592,061,706 on January 8th 2014.

## **Peercoin**

Unlike Bitcoin or Litecoin, which are cryptocurrencies based on a proof-of-work implementation, Peercoin or PPC uses a hybrid proof-of-stake and proof-of-work system. It uses the same source code as Bitcoin for its network design and is currently the 4th largest cryptocurrency (market cap of USD\$ 120,985,892). The system doesn't limit the number of coins that can be generated but ensures a one percent annual rate of inflation. Along with this feature, the mining process is considered more energy efficient which makes it scalable in the long term period.